Derivation Note B2.103

Recursive Labor Allocation (A1.12): Decomposing B1.20 Recursive Paper 1 Commitments by Formalizing How A1.12 Labor Allocation Applies Recursively With Level-Appropriate Labor — LLMs as Labor at Cell Scope, Cells as Labor at Aspect Scope, Aspects as Labor at Self Scope, While Human Governance Remains Constant at Each Level

This work derives from and formalizes the instinct/reasoning separation pattern introduced in "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026), the second paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026).

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Abstract

Paper 2's commitment B1.20 — that CKS Selves exhibit six points of architectural advantage over biological analogs — specifies that all Paper 1 architectural commitments apply recursively at cell, aspect, and Self scope. This note formalizes the recursive application of commitment A1.12, the labor allocation framework. A1.12 distinguishes governance (human) from labor (performable by LLMs or lower-level entities) at the scope of a single coordination cell. Recursive A1.12 extends that distinction to all three structural levels: at cell scope, LLM instinct-layer operations are labor while humans govern through cell DNA; at aspect scope, cell execution of coordination work is labor while humans govern through aspect DNA; at Self scope, aspect execution of integration work is labor while humans govern through Self DNA. The result is a nested labor structure — LLM labor within cell labor within aspect labor within Self labor — with human governance applying independently at each level rather than nesting in the same way. What is constant across levels: the governance-labor distinction itself. What differs: which entity performs labor. The nested labor structure enables operational efficiency at every scope without requiring human execution work at any scope; human governance is exercised through DNA authoring at each level, not through performing the work the DNA governs.

1. Why Recursive Labor Allocation Needs to Be Formalized as a Standalone Operational Variant

B2.103 is the sixth of thirteen notes decomposing B1.20's recursive Paper 1 commitments. Notes B2.98 through B2.102 have addressed cross-lineage mating compatibility, structural evolvability on operational timescales, reversibility, and recursive governance (A1.01), among the preceding items in the decomposition series. B2.103 takes up commitment A1.12.

A1.12 in its original Paper 1 formulation distinguishes three modes by which coordination work can be performed: direct human labor, LLM labor under orchestration rules, and stable-cell automation. It establishes that the mode of labor does not condition the authority structure. The human-governed note's operational test requires that all three modes remain available, that humans retain inspect/modify/override rights over substrate content regardless of mode, and that work can move between modes without changing the content's authority status or structure.

That original formulation operates at the scope of a single coordination cell. Paper 2's three-level structure — cell, aspect, Self — raises a question B1.20 answers directly: do Paper 1's commitments hold at every level, including aspect and Self scope? The answer is yes. B2.103's task is to formalize what that yes means specifically for A1.12. The claim is precise: labor allocation does not merely re-apply at each level unchanged; it re-applies with level-appropriate labor actors, producing a nested labor structure that Paper 1's single-cell A1.12 did not and could not specify.

Naming this recursive application as a separate operational variant matters for the prior-art series. The nested labor structure — LLMs as labor within cells, cells as labor within aspects, aspects as labor within Selves — is a patentable architectural commitment. Its derivation from A1.12 plus B1.20 is what places it in the CKS prior-art chain rather than as a separate invention. Each note in the series reduces the territory within which any party can claim novel invention without encountering existing prior art; B2.103 reduces the specific territory covering multi-level nested labor allocation with independent per-level human governance as an architectural pattern for governed AI systems.

2. The Recursive Application: Three Scopes, One Constant, Three Labor Actors

A1.12 at Cell Scope

The cell scope is where Paper 1's A1.12 formulation directly applies without modification. The LLM performs cell operational work: it consults cell DNA rules, processes inputs, and produces outputs. This is labor at cell scope — the execution of cell operations. Humans govern cell behavior through cell DNA: they author orchestration rules, define schemas, set conflict-handling policies, and retain inspect/modify/override rights over all substrate content. Cell DNA authoring is governance; LLM instinct-layer operation is labor. The governance-labor distinction at cell scope is precisely what A1.12's Paper 1 formulation specifies.

A1.12 at Aspect Scope

The aspect scope is Paper 2's first level of structural extension. An aspect is a coordination arrangement of cells serving a particular purpose; cells perform coordination labor within the aspect. Cells execute the tasks that constitute aspect coordination; cell outputs contribute to aspect outcomes; the work of executing the aspect's coordination purpose is performed by cells. Humans govern aspect coordination through aspect DNA: they author orchestration substrates that define how cells in the aspect relate to one another, what the aspect's coordination purpose is, and how cell outputs compose into aspect-level results. The governance decision — how aspects should coordinate — is human. The labor — executing coordination through cell operations — is cells.

A structural observation is important at aspect scope: LLMs are labor-within-labor here. LLMs perform cell operations as cell-level labor; cells perform aspect coordination as aspect-level labor. The same LLM-under-rule instinct operations that constitute labor at cell scope are components of what constitutes labor at aspect scope. This is not circular — it is hierarchical. The hierarchy is: LLM execution is nested within cell operations, which are nested within aspect coordination. The nesting is operational, not architectural collapse; the cell scope and aspect scope remain distinct levels with their own DNA layers and their own governance.

A1.12 at Self Scope

The Self scope is Paper 2's second level of structural extension. The Self is the integrated whole that holds multiple aspects as facets of one CKS-governed intelligence; aspects — and through them, cells — perform Self integration labor. Aspects contribute to Self-level integration outcomes; the work of executing the Self's integrated intelligence function is performed by aspects. Humans govern Self integration through Self DNA: they author orchestration substrates that specify how aspects compose, how institutional knowledge accumulates at Self scope, and how the Self functions as a unified whole under governance. The governance decision — how the Self should integrate — is human. The labor — executing integration through aspect operations — is aspects. Self DNA authoring is governance; aspect execution of integration work is labor.

The labor-within-labor structure extends one level further at Self scope: LLMs perform cell operations as cell-level labor, cells perform aspect coordination as aspect-level labor, and aspects perform Self integration as Self-level labor. Three tiers of labor nest within one another; human governance applies independently at each tier through level-appropriate DNA, without itself nesting in the same way.

What Is Constant Across All Three Scopes

The governance-labor distinction holds at every scope: governance is human, labor is non-human or lower-level. Humans never perform labor in the AI operational sense at any level — they govern. The governing act is DNA authoring: writing orchestration rules, schemas, conflict-handling policies, and integration substrates at the appropriate level. The labor act is executing work under those rules. The constant across cell, aspect, and Self scope is A1.12's core commitment: governance and labor are distinct, governance is human, and labor can be performed by non-human or lower-level entities without affecting the authority structure.

What Differs Across Scopes

The labor actor differs at each level: at cell scope, LLM instinct operations are labor; at aspect scope, cell execution is labor; at Self scope, aspect execution is labor. The work performed by labor differs correspondingly: cell-scope labor is processing inputs under DNA rules; aspect-scope labor is executing coordination across cells for a purpose; Self-scope labor is performing integration across aspects into unified whole function. The scope-appropriate labor actors and work constitute the level-specific instantiation of A1.12.

3. What Makes Recursive Labor Allocation Architecturally Distinctive

Conventional AI architectures do not typically carry an explicit multi-level labor allocation architecture. The standard pattern is binary: a model executes, and humans deploy and monitor. The deployment and monitoring layer is not formalized as a labor allocation structure at multiple nested scopes; the model's execution is not distinguished from governance over that execution at each level of a structural hierarchy. The result is that labor and governance remain conflated — the question of who performs what at which scope is not a named architectural commitment.

CKS recursive A1.12 creates a three-tier nested labor structure with independent human governance at each tier. The tiers are not merely functional layers; they are scopes at which the governance-labor distinction independently applies. Any deployment implementing the three-level structure carries three independent governance scopes, each with its own DNA layer, its own human authority, and its own labor actor. A failure of labor at one scope — a cell failing to execute correctly — does not constitute a failure of governance at that scope; the governance architecture is independent of the labor architecture at each level.

The architectural distinctiveness is the combination: nested labor (LLM within cell within aspect within Self) with non-nested, independent governance (human authority applied separately at each level through level-appropriate DNA). Nested labor without governance independence would produce cascading authority vulnerabilities — problems at one scope propagating unchecked through the hierarchy. Non-nested governance without labor nesting would require humans to perform execution work at every level, which does not scale. The specific combination — nested labor with independently applied governance at each scope — is what makes recursive A1.12 an architectural commitment rather than an implementation convenience.

4. The Biological Analog as Conceptual Scaffold

Recursive labor allocation parallels nested metabolic labor in biological systems. Molecules perform biochemical work within cells: enzymatic reactions, energy conversion, signal transduction. Cells perform functional work within tissues: secretion, contraction, signal propagation. Tissues perform physiological work within organs: filtration, circulation, gas exchange. The labor at each level is performed by entities at that level; higher-level entities do not perform lower-level work directly. A kidney does not perform enzymatic reactions; enzymatic reactions in kidney cells contribute to filtration at the organ level.

The CKS recursive A1.12 structure follows this pattern: LLMs perform instinct-layer operations, not aspect coordination directly; aspects perform Self integration, not cell-level processing directly. The labor is nested; higher-level labor is constituted by lower-level labor without the higher-level entity performing lower-level work itself.

Where CKS exceeds the biological analog is precisely on the governance dimension. Biology has no equivalent of level-appropriate human governance independent at each scope. Enzymatic reactions are not governed by deliberate human authority structures; metabolic organization emerges through evolutionary processes rather than through explicit governance decisions. CKS recursive A1.12 is the governed architectural analog: the nested labor structure that biology exhibits, combined with independent human governance at every level through level-appropriate DNA authoring. B1.20's list of points at which CKS exceeds biology includes this governed-labor-nesting combination as an architectural advantage; B2.103 is its per-commitment formalization.

5. Inherited Paper 1 Commitments

A1.12 itself is the commitment being formalized. Paper 2's recursive application does not replace the single-cell A1.12 formulation; it extends it to the aspect and Self scopes Paper 1 did not address. The cell-scope formulation from Paper 1 remains unchanged and is the foundation on which recursive extension builds. The extension names precisely what recursive application requires: level-appropriate labor actors at each additional scope.

A1.01 (human-governed, authority not labor) holds at every level. The human-governed commitment specifies three rights — inspect, modify, override — over substrate content at all times. Recursive A1.12 requires those rights to hold at all three scopes independently: humans hold inspect/modify/override rights over cell DNA, aspect DNA, and Self DNA separately. The governance-labor distinction that A1.01 anchors extends through every scope without modification.

A2.04 (rule authoring as governance mechanism) is the governance mechanism at each level. DNA authoring — the act of writing orchestration rules, schemas, and coordination substrates at a given scope — is what human governance means operationally at cell, aspect, and Self scope. The rule-authoring commitment from Paper 1 holds at all three levels; Paper 2 specifies that each level has its own DNA layer, meaning rule authoring applies independently at each scope. B2.102's recursive governance note formalizes A1.01's recursive application; B2.103 formalizes the labor side that A1.01's recursive governance governs.

The nested labor structure is Paper 2's extension of Paper 1's A1.12. The derivation chain is direct and traceable: B1.20 commits to recursive Paper 1 commitments across cell, aspect, and Self scope; A1.12 is a Paper 1 commitment requiring formalization of labor allocation; recursive A1.12 applies A1.12 at each of the three levels with level-appropriate labor actors. No commitments outside Paper 1 and Paper 2 are introduced.

6. Operational Implications

Recognize the nested labor structure in deployments. Self integration labor is aspect execution. Aspect coordination labor is cell execution. Cell operational labor is LLM instinct operation. A deployment that treats LLM execution as the only labor layer — ignoring cell-scope and aspect-scope labor allocation — has not implemented recursive A1.12. Recognition means identifying which entity performs labor at each scope and confirming that humans govern at each scope through level-appropriate DNA authoring.

Human governance is independent at each level. The authority structure for cell DNA is architecturally distinct from the authority structure for aspect DNA or Self DNA. Different humans may hold authority at different scopes; institutional authority structures determine which humans govern which DNA layers. What the architecture requires is that governance is present at each scope, not that the same humans govern all three. B2.102’s recursive governance formalization covers the governance side; B2.103 covers the labor side that governance governs.

Labor allocation principles apply through the full lifecycle. Birth governance per B2.41 applies the labor/governance distinction specifically to origination events; recursive A1.12 covers the full lifecycle, including instinct evolution (B1.10), DNA evolution (B1.11), action-feedback evolution (B1.12), and death (B1.08). At every lifecycle event, the labor actor at each scope performs execution work; humans govern through level-appropriate DNA authoring. Lifecycle-event-specific applications do not override the general framework; they instantiate it at specific event types.

Operational efficiency through labor nesting. The nested labor structure enables operational efficiency at every level: LLM instinct operations execute cell work without human execution involvement; cell operations execute aspect coordination without humans performing cell-level execution; aspect operations execute Self integration without humans performing aspect-level execution. Human governance is exercised through DNA authoring — a structurally distinct act that does not require performing execution labor. The architecture is efficient not because it removes governance but because it separates governance from execution consistently at every scope.

7. Limits

Recursive A1.12 does not make governance automated. Governance is human at every scope. Recursive application does not make governance a mechanical property that propagates downward automatically; it means human governance is present and independently active at each scope. Each level requires actual human authority exercised through DNA authoring, retained inspect/modify/override rights, and governance decisions appropriate to that scope.

The nested labor structure does not mean higher levels govern lower levels directly. Each level has its own governance through its own DNA. Self governance does not subsume cell governance; aspects do not govern cells on behalf of Selves. The independence of governance at each scope is a positive commitment, not a side effect of the three-level structure. Higher levels operate over lower levels as content domain — not as governance authority extending downward to replace lower-level governance.

Labor nesting is operational, not architectural collapse. The fact that cells-as-labor-at-aspect-scope are themselves constituted by LLM-labor-at-cell-scope does not merge the cell and aspect levels into one. The levels remain architecturally distinct — each with its own scope, its own DNA layer, its own human authority, and its own labor actor. The operational nesting of labor within labor describes how execution propagates through the structure; it does not alter the architectural distinctness of the levels.

Recursive A1.12 is the labor allocation commitment; per-scope mechanisms are covered elsewhere in the decomposition series. B2.99 through B2.101 address structural evolvability and related per-level operational mechanisms. B2.104 through B2.110 address the remaining B1.20 decomposition commitments. B2.103 formalizes the labor allocation commitment itself; detailed execution mechanisms at each scope belong to the relevant notes in the series.

8. Operational Test

A system implements recursive labor allocation (A1.12) if and only if all of the following are true:

1. At cell scope, LLM instinct-layer operations perform cell execution work under cell DNA rules authored by humans, and humans hold inspect/modify/override rights over cell DNA content.

2. At aspect scope, cells perform coordination execution work under aspect DNA rules authored by humans, cell-level LLM labor (condition 1) is nested within this coordination execution, and humans hold inspect/modify/override rights over aspect DNA content.

3. At Self scope, aspects perform integration execution work under Self DNA rules authored by humans, aspect-level cell labor (condition 2) is nested within this integration execution, and humans hold inspect/modify/override rights over Self DNA content.

4. Human governance at each of the three scopes is independent: the authority structure for cell DNA is not the authority structure for aspect DNA or Self DNA.

5. No human performs operational execution labor at any of the three scopes — the human role at each scope is DNA authoring and rights retention, not execution.

A system that satisfies condition (1) at cell scope but does not extend the governance-labor distinction to aspect and Self scope has implemented Paper 1's A1.12 but not its recursive Paper 2 extension per B1.20.

9. Naming and Position in the Decomposition Series

B2.103 is the sixth of thirteen notes decomposing B1.20. The first five notes (B2.98–B2.102) have addressed cross-lineage mating compatibility, structural evolvability on operational timescales, reversibility of cell states, and recursive governance (A1.01). B2.103 addresses recursive labor allocation (A1.12). The remaining seven notes will address recursive substrate-as-source-of-truth (A1.08 — B2.104), recursive retraceability (A1.07 — B2.105), recursive composition requirements (A1.13 — B2.106), recursive authority architecture (A2.01–A2.04 — B2.107), recursive conflict-as-first-class (A1.03 — B2.108), recursive operational tests (B2.109), and recursive commitments verification (B2.110), which closes Phase B2.

Each note in the B1.20 decomposition series contributes a per-commitment formalization of what recursive Paper 1 application means. B2.103's contribution is the nested labor structure: LLMs as labor at cell scope, cells as labor at aspect scope, aspects as labor at Self scope, with human governance independent at each level through level-appropriate DNA authoring. Naming this as a standalone prior-art note establishes that the nested-labor-with-independent-governance combination is not a novel invention available for independent claim — it is a derivation from A1.12 and B1.20, publicly dated and attributed to the CKS theory series.

Source Papers

Li, W. (April 2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.*

Li, W. (April 2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.*

Cross-Reference Notes

A1.01 — Human-governed: authority not labor

A1.12 — The labor allocation framework: three modes

A2.04 — Rule authoring as governance mechanism

B1.20 — Where CKS exceeds biology: six points of architectural advantage

B2.41 — Birth governance vs. labor (lifecycle-event-specific application of A1.12)

B2.102 — Recursive governance (A1.01)

B2.104 — Recursive substrate-as-source-of-truth (A1.08) [forthcoming]